

Seminar 10

NP-completeness

1. Prove that the decision MIS - Maximum Independent Set problem is NP-hard. Hint: Reduce 3-SAT to Maximum Independent Set.
2. Prove that the CLIQUE decision problem is NP-complete. Hint: reduce MIS (Maximum Independent Set) to CLIQUE.
3. Prove that the halting problem is NP-hard.
4. What is the decision problem corresponding to the knapsack problem?
5. Prove that the discrete knapsack problem is NP-hard.
6. Prove that the Vertex Cover decision problem is NP-complete. Hint: Reduce MIS (Maximum Independent Set) to Vertex Cover.
7. Prove that the PARTITION problem is (weakly) NP-hard (given a set of numbers, can it be split into two subsets of equal sum?).
8. Prove that the PARTITION problem is (weakly) NP-hard (given a set of numbers, can it be partitioned into two disjoint subsets of equal sum?).